

|                                                |                                       |
|------------------------------------------------|---------------------------------------|
| Program : <b>Diploma in Textile Technology</b> |                                       |
| Course Code : <b>3069</b>                      | Course Title: <b>Woven Design Lab</b> |
| Semester : <b>3</b>                            | Credits: <b>No Credit</b>             |
| Course Category: <b>Program core</b>           |                                       |
| Periods per week: <b>4 (L:1, T:0, P:3)</b>     | Periods per semester: <b>60</b>       |

### Course Objectives:

- To impart the basic knowledge on woven fabrics and development of designs for the construction of fabric.

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                  | Duration (Hours) | Cognitive level |
|-----------------|------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Understand the basic classification of fabrics.                              | 10               | Understanding   |
| CO2             | Prepare interlacement diagram and various ornamentations of woven fabrics.   | 16               | Applying        |
| CO3             | Understand the basic elements of textile designs & Fabric sample development | 17               | Applying        |
| CO4             | Practicing CATD system and develop patterns and designs on paper             | 14               | Applying        |
|                 | Lab Exam                                                                     | 3                |                 |

### CO-PO Mapping:

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| <b>CO1</b>      | 3   |     |     |     |     |     |     |
| <b>CO2</b>      | 3   |     |     |     |     |     |     |
| <b>CO3</b>      | 3   |     |     |     |     |     |     |
| <b>CO4</b>      | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b> | <b>Description</b>                                                                                                   | <b>Duration (Hours)</b> | <b>Cognitive level</b> |
|------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>Understand the basic classification of fabrics used.</b>                                                          |                         |                        |
| M1.01                  | Classification of fabrics – Woven, Non-woven, Knitted and Braided fabrics.                                           | 4                       | Understanding          |
| M1.02                  | Identify warp and weft threads, selvedge, weaving method and machine required for producing the fabric.              | 6                       | Understanding          |
| <b>CO2</b>             | <b>Construct interlacement diagram and various ornamentations of woven fabric</b>                                    |                         |                        |
| M2.01                  | Interlacement diagram – Warp way and weft way                                                                        | 2                       | Understanding          |
| M2.02                  | Graphical representation of structure of a woven design.                                                             | 4                       | Applying               |
| M2.03                  | Ornamentation of elementary weaves- Structural ornamentation & Decorative ornamentation                              | 4                       | Understanding          |
| M2.04                  | Decorative ornamentation - by using colour – stripe, cross over, check patterns)- by using yarns of different count. | 6                       | Applying               |
|                        | Lab Exam - I                                                                                                         | 1.5                     |                        |
| <b>CO3</b>             | <b>Understand the basic elements of textile designs and construct samples</b>                                        |                         |                        |
| M3.01                  | Understanding of elements of Textile Design. Draw Design, Draft, Peg plan, Prepare Tie-up and Treadling plan.        | 4                       | Applying               |
| M3.02                  | Practice Drawing– in and denting for elementary Designs & understand the role of healds & reeds.                     | 3                       | Applying               |
| M3.03                  | Practice the construction of fabrics of different Designs using table looms or sample looms                          | 6                       | Applying               |
| M3.04                  | Extracting fundamental details like ends and picks per unit space (inch), count of warp and weft.                    | 4                       | Applying               |
| <b>CO4</b>             | <b>Fabric sample development and Understand the use of CATD system.</b>                                              |                         |                        |
| M4.01                  | Understand the development of Textile designs using CATD and prepare draft, pegplan by CATD                          | 4                       | Understanding          |

|       |                                                                                                                                                                     |     |               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|
| M4.02 | Preparation of stripe & check patterns with different colour combinations using CATD & collect different samples on paper.(print outs) Textiles with their details. | 5   | Applying      |
| M4.03 | Develop dobby & Jacquard figures using CATD System.                                                                                                                 | 5   | Understanding |
|       | Lab Exam - II                                                                                                                                                       | 1.5 |               |

The students should keep album with samples and submit certified & bonafide record at the time of Fabric Manufacture practical examination.

#### **Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                         |
|------------|--------------------------------------------------|
| T1         | Walton's Advanced Textile Design by Z. Crosiciki |
| R1         | Structural Fabric Design by James W. Klibbe      |
| R2         | Woven cloth construction by R. Mark              |
| R3         | Grammar of Textile Design by H. Nisbet           |
| R4         | Woven structure and Design by Dori Goernar       |

#### **Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                         |
|--------------|-----------------------------------------------------------------------------|
| 1            | <a href="http://www.youtube.com">www.youtube.com</a>                        |
| 2            | <a href="https://textilelearner.net/">https://textilelearner.net/</a>       |
| 3            | <a href="https://www.fibre2fashion.com/">https://www.fibre2fashion.com/</a> |